

# The Telegraph Model of Gene Expression: A Triangular Cross-Validation of Stochastic Simulation

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## Abstract

Gene expression is a stochastic process: even genetically identical cells in the same environment produce very different numbers of messenger RNA (mRNA) molecules. The telegraph model explains the main source of this variability. It treats a gene as a two-state (ON/OFF) switch that transcribes mRNA only while active. We study the telegraph model through three algebraically equivalent descriptions, all following from the same chemical master equation, and check that their independent implementations agree: we (i) generate stochastic trajectories with the Gillespie stochastic simulation algorithm (SSA); (ii) derive and solve a closed system of ordinary differential equations (ODEs) for the first and second moments; and (iii) draw independent samples directly from the analytic steady-state distribution, using the classic Beta–Poisson law together with an explicit Bernoulli gene state. The main contribution is the implementation cross-validation that links the three. The code is available at <https://github.com/zakhar-guzii/telegraph-model-gene-expression>.

## 1 Introduction

Gene expression is variable. Two cells with the same DNA, grown together in the same environment, can have different numbers of copies of the same mRNA or protein [6, 7]. This difference between cells is called phenotypic noise. It happens naturally because of random molecular interactions. Experimentally this variability separates into two parts [6]: *extrinsic* noise, caused by differences between cells—their size, cell-cycle stage, or the pool of available polymerases—and *intrinsic* noise, the randomness of the transcription and degradation events themselves, which persists even between two identical copies of a gene inside the same cell.

Inside a cell, the key players—a single copy of the gene, a handful of transcription factors, and a few dozen mRNA molecules—are present in such small numbers that the discreteness and randomness of individual chemical events cannot be averaged away. The telegraph model studied here is the simplest framework that describes this intrinsic component. It treats the promoter as a two-state device that randomly switches between an active (ON) and an inactive (OFF) state [8]. While ON, it produces mRNA molecules through transcription in random bursts; each molecule is then degraded at a random time. As a result, the mRNA count is a discrete-valued stochastic process.

Despite its simplicity, the telegraph model reproduces the bursty, *over-dispersed* mRNA distributions seen in single-molecule experiments, which count individual mRNAs inside single cells [5],

and it can be analysed exactly [4]. A convenient way to measure this spread is the *Fano factor*  $F = \sigma^2/\mu$  [9], the ratio of the variance to the mean of the mRNA count. Pure Poisson production and decay, in which every event is independent, has variance equal to mean, giving the *Poisson baseline*  $F = 1$ . “Over-dispersed” means  $F > 1$ : the random switching of the gene adds extra variability on top of that baseline.

We treat the telegraph model as a case study in how a stochastic process can be described from several mathematically distinct viewpoints. We implement three approaches that look unrelated:

1. a stochastic simulation (the SSA), which follows a single cell through time;
2. a system of moment ODEs, which gives the population mean and spread deterministically;
3. a steady-state sampler, which describes the population after a long time.

Each is correct on its own, and the three are algebraically equivalent: all follow from the same chemical master equation. What differs is the implementation. The three share no code and fail in different ways—an error in the event loop, a mis-derived moment equation, or a wrong stationary law would each disturb one description while leaving the others intact—so agreement between them is evidence against implementation defects. It is not evidence about the master equation itself, which all three inherit.

## 2 Model Formulation

The state of the system at time  $t$  is the pair

$$(G(t), R(t)), \quad G(t) \in \{0, 1\}, \quad R(t) \in \mathbb{N}, \quad (1)$$

where the binary variable  $G$  records whether the gene is OFF ( $G = 0$ ) or ON ( $G = 1$ ), and the integer  $R$  counts the number of mRNA molecules present. Four basic reactions define the dynamics of the system. Each reaction has a *propensity*  $a_j$ , the rate at which it fires at any given moment:

| Reaction | Description       | Effect                | Propensity                   |
|----------|-------------------|-----------------------|------------------------------|
| $R_1$    | Gene activation   | $G : 0 \rightarrow 1$ | $a_1 = k_{\text{on}}(1 - G)$ |
| $R_2$    | Gene inactivation | $G : 1 \rightarrow 0$ | $a_2 = k_{\text{off}} G$     |
| $R_3$    | RNA synthesis     | $R \rightarrow R + 1$ | $a_3 = k_{\text{syn}} G$     |
| $R_4$    | RNA degradation   | $R \rightarrow R - 1$ | $a_4 = k_{\text{deg}} R$     |

All four rate constants are positive. The switching rates  $k_{\text{on}}, k_{\text{off}}$  set how quickly the promoter turns on and off, and their ratio fixes the long-run fraction of time the gene is active; the factor  $G$  in  $a_3$  shuts transcription off entirely when the gene is OFF; and  $k_{\text{deg}}$  is a per-molecule decay rate, so the total degradation propensity  $k_{\text{deg}} R$  grows with the amount of mRNA present. Together the rates  $(k_{\text{on}}, k_{\text{off}}, k_{\text{syn}}, k_{\text{deg}})$  and an initial condition  $(g_0, r_0)$  at time  $t_0$  fully determine the law of the process.

**Burst size.** Once the gene is ON, the only reaction that can switch it off fires at the constant rate  $k_{\text{off}}$ , so the duration of an active period is exponential with mean  $1/k_{\text{off}}$ ; during that period transcription fires at rate  $k_{\text{syn}}$ . The mean number of transcripts made per active period—the *burst size*  $B$ —is therefore

$$B = k_{\text{syn}} \mathbb{E}[\text{ON duration}] = \frac{k_{\text{syn}}}{k_{\text{off}}}. \quad (2)$$

Bursts of this kind are observed directly in single cells [10, 5], and the exact consequences of promoter switching for the mRNA distribution, including the limits recovered in Section 4.2, are derived in [11].

Because each waiting time is exponential and depends only on the current state,  $(G(t), R(t))$  is a continuous-time Markov chain (CTMC). Its distribution  $P_t(g, r) = \Pr\{G(t) = g, R(t) = r\}$  evolves according to the chemical master equation [13, 14, 15, 16]

$$\begin{aligned} \frac{d}{dt}P_t(g, r) = & k_{\text{on}}[\delta_{g,1}P_t(0, r) - \delta_{g,0}P_t(g, r)] \\ & + k_{\text{off}}[\delta_{g,0}P_t(1, r) - \delta_{g,1}P_t(g, r)] \\ & + k_{\text{syn}}\delta_{g,1}[P_t(1, r-1) - P_t(1, r)] \\ & + k_{\text{deg}}[(r+1)P_t(g, r+1) - rP_t(g, r)], \end{aligned} \quad (3)$$

with the convention  $P_t(g, -1) = 0$ . Equation (3) is the basis for all three approaches below: the SSA samples trajectories whose distribution solves it, the moment ODEs are derived from it exactly, and the steady-state sampler draws from its  $t \rightarrow \infty$  fixed point.

## 3 Methods

### 3.1 Stochastic Simulation (Gillespie SSA)

The *stochastic simulation algorithm* (SSA) of Gillespie [1, 2, 3] generates statistically exact sample paths of a CTMC, making no approximation to the master equation (3). Its key idea is that, given the current state, two questions separate: when does the next reaction occur, and which reaction is it? With  $a_0 = \sum_{j=1}^4 a_j$  the *total propensity*, the waiting time is  $\tau \sim \text{Exp}(a_0)$  and reaction  $j$  fires with probability  $a_j/a_0$ .

Our implementation draws these by inversion, using two independent uniforms  $r_1, r_2 \sim \text{U}(0, 1)$  per step from a seeded `numpy.random` generator [17], one seed per experiment:

$$\tau = \frac{1}{a_0} \ln \frac{1}{r_1}, \quad j = \min \left\{ k : \sum_{i=1}^k a_i > r_2 a_0 \right\}. \quad (4)$$

The time is then advanced by  $\tau$ , the state updated by reaction  $j$ , the reaction propensities recomputed, and the loop repeats for a fixed number of events  $n_{\text{sim}}$ . If the total propensity reaches  $a_0 = 0$ , no reaction can fire, so the state is absorbing and is held fixed afterwards.

A single trajectory is the history of one cell. To obtain population statistics, we run  $N_{\text{rep}}$  independent trajectories with identical parameters and average over them. Each trajectory advances by its own random time increments, so the same event index occurs at different times in different trajectories. We therefore resample every trajectory onto a shared uniform time grid by a zero-order hold before averaging; the state is piecewise constant between events, so this introduces no error. The empirical moments are then ensemble averages at each grid time, for example

$$\widehat{\mu}_R(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i(t), \quad \widehat{\mu}_G(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} G_i(t), \quad (5)$$

with sample variances and the sample gene–RNA covariance defined analogously.

**Simulation horizon.** The loop counter  $n_{\text{sim}}$  bounds the number of *events*, not the elapsed time, so an ensemble has no common end time. We therefore take the shared grid no further than the earliest-finishing trajectory,  $t_{\text{end}} = \min_i t_i^{\text{last}}$ : extending past that point would freeze the exhausted trajectories at their final state and bias the ensemble toward whatever those particular cells happened to be doing when they ran out of events. Each figure therefore ends at its own, seed-dependent time, in every case a large multiple of the relaxation time. This rule is not entirely neutral: a trajectory exhausts its event budget fastest when its total propensity is high, so  $t_{\text{end}}$  is itself weakly correlated with the ensemble state, and we do not attempt to quantify the resulting bias here.

Figure 1 shows a typical sample path generated by the algorithm. Averaging many such trajectories according to (5) gives the ensemble moments that Section 4.3 compares with the moment ODEs of Section 3.2.

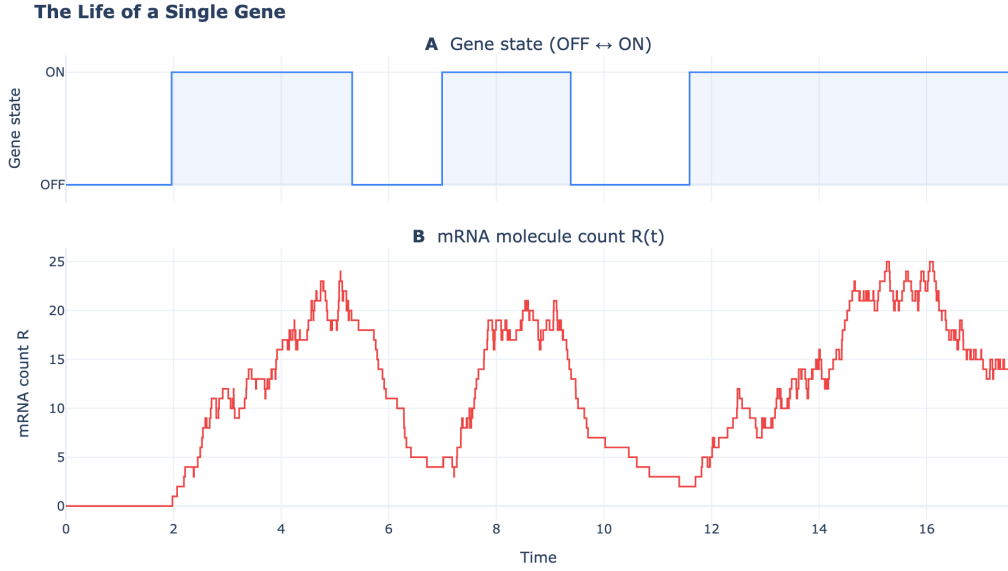


Figure 1: A single SSA sample path at the reference parameters  $k_{\text{on}} = k_{\text{off}} = 0.5$ ,  $k_{\text{syn}} = 20$ ,  $k_{\text{deg}} = 1$ . Top: the gene state switches between OFF and ON at random times. Bottom: the mRNA count rises in bursts during ON periods and decays between them. Averaging many such trajectories gives the sample moments used below.

### 3.2 Moment ODEs

The SSA describes individual cells. Often we instead want deterministic equations for population-level statistics—the *moments*. A moment is an expectation: the mean  $\mu_R = \mathbb{E}[R]$  is the average mRNA level, the variance  $\sigma_R^2 = \mathbb{E}[R^2] - \mathbb{E}[R]^2$  measures the noise, and the covariance  $C_{RG} = \mathbb{E}[RG] - \mathbb{E}[R]\mathbb{E}[G]$  measures how gene activity and mRNA level vary together.

Applying the standard moment rule for a chemical master equation—the expected rate of change of any observable is the sum over reactions of each propensity times the change it makes—to

$f = G, R, R^2, GR$  in turn yields four coupled equations for the telegraph model [16, 15]:

$$\frac{d\mu_G}{dt} = k_{\text{on}}(1 - \mu_G) - k_{\text{off}}\mu_G, \quad (6)$$

$$\frac{d\mu_R}{dt} = k_{\text{syn}}\mu_G - k_{\text{deg}}\mu_R, \quad (7)$$

$$\frac{d\sigma_R^2}{dt} = 2k_{\text{syn}}C_{RG} - 2k_{\text{deg}}\sigma_R^2 + k_{\text{syn}}\mu_G + k_{\text{deg}}\mu_R, \quad (8)$$

$$\frac{dC_{RG}}{dt} = k_{\text{syn}}\sigma_G^2 - (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})C_{RG}. \quad (9)$$

Each has a direct reading. Activation pushes mean activity toward 1 and inactivation pulls it toward 0; mRNA is made in proportion to the active fraction  $\mu_G$  and removed in proportion to the amount present; the mRNA variance is driven up by the coupling to gene activity (the  $C_{RG}$  term) and by the randomness of production and decay, and down by degradation; and the covariance is created by transcription passing the gene's own fluctuations  $\sigma_G^2$  on to the mRNA, then damped at the combined rate of switching and degradation.

**Exact closure.** In a general nonlinear reaction system the equation for each moment depends on higher moments, so the system never closes and one has to add approximate closure schemes [12, 16, 15]. Here it closes *exactly*. The only possible nonlinearity is the variance of the gene state, but because  $G$  is binary,

$$\sigma_G^2 = \mathbb{E}[G^2] - \mu_G^2 = \mathbb{E}[G] - \mu_G^2 = \mu_G(1 - \mu_G), \quad (10)$$

which depends on the mean  $\mu_G$  alone. Putting (10) into (9) turns the four equations (6)–(9) into a *closed, exact* linear system in  $(\mu_G, \mu_R, \sigma_R^2, C_{RG})$ , whose solutions are the true moments of the master equation (3). A four-variable numerical integration is therefore enough, with no closure approximation anywhere in the code.

**Implementation.** The four right-hand sides (6)–(9), with  $\sigma_G^2$  substituted by (10), are coded directly and passed to the explicit Runge–Kutta solver `scipy.integrate.solve_ivp` [18] (method RK45, the embedded Dormand–Prince pair [19]) from the initial conditions  $\mu_G(t_0) = g_0$ ,  $\mu_R(t_0) = r_0$ ,  $\sigma_R^2(t_0) = C_{RG}(t_0) = 0$ , evaluated on a uniform grid of 1000 time points. Section 4.3 overlays them on the SSA ensemble at the same rates and the same initial condition.

**Steady-state Fano factor.** Setting the right-hand side of (9) to zero gives  $C_{RG}^{ss} = k_{\text{syn}}\sigma_G^2/(k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})$  with  $\sigma_G^2 = \mu_G^{ss}(1 - \mu_G^{ss})$  by (10); substituting that into (8) at stationarity, where  $2k_{\text{deg}}\sigma_R^2 = 2k_{\text{syn}}C_{RG} + k_{\text{syn}}\mu_G + k_{\text{deg}}\mu_R$  and  $k_{\text{syn}}\mu_G = k_{\text{deg}}\mu_R$  by (7), and dividing by  $\mu_R^{ss}$  yields the Fano factor in closed form:

$$F^{ss} = \left. \frac{\sigma_R^2}{\mu_R} \right|_{ss} = 1 + \frac{k_{\text{syn}}k_{\text{off}}}{(k_{\text{on}} + k_{\text{off}})(k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})}. \quad (11)$$

The additive 1 is the Poisson baseline: it is what survives when  $k_{\text{off}} \rightarrow 0$  and the gene stops switching, so the mRNA count is a pure birth–death process. The second term is the excess contributed by promoter switching, and it vanishes in two distinct ways—as  $k_{\text{off}} \rightarrow 0$  (the gene is always ON) and as  $k_{\text{on}}, k_{\text{off}} \rightarrow \infty$  at fixed ratio, where a promoter that switches very fast averages out.

Setting the time derivatives in (6)–(7) to zero in the same way gives the exact stationary means used as the reference throughout Section 4:

$$\mu_G^{ss} = \frac{k_{\text{on}}}{k_{\text{on}} + k_{\text{off}}}, \quad \mu_R^{ss} = \frac{k_{\text{syn}}}{k_{\text{deg}}} \mu_G^{ss}. \quad (12)$$

### 3.3 Steady-State Sampler

As  $t \rightarrow \infty$  the telegraph model forgets its initial condition and converges to a unique stationary distribution  $P_\infty(g, r)$ . Instead of simulating a long transient to reach it, we can sample from  $P_\infty$  directly. Peccoud and Ycart [4] showed that at steady state the mRNA count follows the classic *Beta-Poisson* law: it is Poisson, but with a mean that is itself random from cell to cell. Let the *mixing fraction*  $p \in [0, 1]$  be the long-run share of time a cell's gene spends ON. Across a population  $p$  is Beta-distributed, and given  $p$  the mRNA count is Poisson with mean  $\lambda = cp$ , where  $c = k_{\text{syn}}/k_{\text{deg}}$  is the *transcription efficiency coefficient*: the mean mRNA level a permanently active gene would reach. With the switching rates  $a = k_{\text{on}}/k_{\text{deg}}$  and  $b = k_{\text{off}}/k_{\text{deg}}$  (so  $k_{\text{deg}}$  fixes the unit of time), the stationary mRNA law is therefore

$$R \sim \text{Poisson}(cp), \quad p \sim \text{Beta}(a, b). \quad (13)$$

Averaging the Poisson kernel over the Beta density gives  $P_\infty(r)$  in closed form, as a confluent hypergeometric expression [4]. We do not need it: because the mixture (13) *is* that distribution, sampling the mixture samples  $P_\infty$  exactly, at the cost of three vectorised draws per count and with no discretisation of the state space.

Our sampler draws the gene state as an explicit Bernoulli( $p$ ) so we can also estimate the joint moments of  $(G, R)$ . For each of the  $N_{\text{rep}}$  samples it draws

$$p \sim \text{Beta}(a, b), \quad f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} p^{a-1}(1-p)^{b-1}, \quad (14)$$

$$G \mid p \sim \text{Bernoulli}(p), \quad \Pr(G = 1 \mid p) = p, \quad \Pr(G = 0 \mid p) = 1 - p, \quad (15)$$

$$R \mid p \sim \text{Poisson}(\lambda), \quad \lambda = \frac{k_{\text{syn}}}{k_{\text{deg}}} p. \quad (16)$$

The key point is that the mRNA count  $R$  is conditioned on the mixing fraction  $p$ , *not* on the current gene state  $G$ . Given  $p$ , the gene state and the mRNA count are drawn independently, so a cell can have  $G = 0$  and still  $R > 0$ —as in a real cell, where mRNA made during the last active period survives past the switch to OFF.

**The joint construction.** Peccoud and Ycart [4] derived the *marginal* law of  $R$ , whereas the checks below need  $\widehat{\sigma}_G^{ss}$  and  $\widehat{C}_{RG}^{ss}$ , which are properties of the joint law. Coupling  $G$  to the same mixing fraction  $p$  that drives the Poisson mean supplies one, and it reproduces the covariance that the master equation gives independently. That reference value follows from setting (9) to zero, as in the Fano-factor derivation of Section 3.2:

$$C_{RG}^{ss} = \frac{k_{\text{syn}} \sigma_G^2}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}} = \frac{k_{\text{syn}} \mu_G^{ss} (1 - \mu_G^{ss})}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}}. \quad (17)$$

Computing the same quantity from the sampler instead gives  $\mathbb{E}[RG] = \mathbb{E}[\mathbb{E}[R \mid p] \mathbb{E}[G \mid p]] = c \mathbb{E}[p^2]$ . Since  $\mathbb{E}[R]\mathbb{E}[G] = c \mathbb{E}[p]^2$ , it follows that  $C_{RG} = c \text{Var}(p)$ . Using the Beta variance  $\text{Var}(p) = ab/((a+b)^2(a+b+1))$  and substituting  $ab/(a+b)^2 = \mu_G^{ss}(1 - \mu_G^{ss})$ ,  $a+b+1 = (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})/k_{\text{deg}}$ , and  $c = k_{\text{syn}}/k_{\text{deg}}$ , this expands to

$$C_{RG} = \frac{k_{\text{syn}}}{k_{\text{deg}}} \cdot \frac{\mu_G^{ss}(1 - \mu_G^{ss}) k_{\text{deg}}}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}} = \frac{k_{\text{syn}} \mu_G^{ss} (1 - \mu_G^{ss})}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}}, \quad (18)$$

which matches (17) term for term. The construction therefore reproduces the joint moments of the master equation. This is an algebraic identity rather than an empirical result: the numerical agreement reported in Section 4.4 confirms the Monte Carlo estimator, not the construction itself.

**Boundary behaviour.** The construction also behaves as expected at the edges of the parameter space. As the switching rates become very slow compared with degradation ( $a, b \rightarrow 0$ ), the  $\text{Beta}(a, b)$  density concentrates its mass at the endpoints  $p = 0$  and  $p = 1$ , and the mixture separates into its building blocks: a single Bernoulli trial for the gene state and, when ON, an ordinary Poisson count for the mRNA. At  $k_{\text{off}} = 0$  the gene never switches off ( $p \equiv 1$ ) and the stationary law reduces to  $\text{Poisson}(k_{\text{syn}}/k_{\text{deg}})$ ; Section 4.4 tests this numerically. At  $k_{\text{deg}} = 0$  no steady state exists and the sampler is undefined.

The joint stationary law is the average of the conditional distributions over the mixing fraction,  $P_{\infty}(g, r) = \int_0^1 \Pr(G = g \mid p) \Pr(R = r \mid p) f(p) dp$ ; drawing  $N_{\text{rep}}$  triples  $(p, G, R)$  and averaging gives unbiased Monte Carlo estimates of the steady-state moments  $\widehat{\mu}_G^{ss}, \widehat{\mu}_R^{ss}$ , and so on.

## 4 Experiments and Results

Unless stated otherwise, the experiments use the symmetric *reference parameters*  $k_{\text{on}} = 0.5$ ,  $k_{\text{off}} = 0.5$ ,  $k_{\text{syn}} = 20.0$ ,  $k_{\text{deg}} = 1.0$ , with  $N_{\text{rep}} = 1000$  trajectories and initial condition  $(g_0, r_0) = (0, 0)$ , giving  $\mu_G^{ss} = 0.5$  and  $\mu_R^{ss} = 10$  by (12). The parameter-exploration runs of Section 4.2 are the one exception: they vary rates around a slower base of  $k_{\text{on}} = k_{\text{off}} = 0.1$ , stated with each run in Table 1. All three parts of the validation are implemented in Python; see the repository linked above.

### 4.1 The Validation Triangle

The three descriptions of Section 3 must agree wherever their domains overlap. We state those overlaps as the three edges of a *validation triangle* and test each one numerically below.

**Edge 1 — SSA  $\leftrightarrow$  Moment ODEs (finite time).** The empirical averages (5) are sample means of independent, identically distributed trajectories, so by the Law of Large Numbers they converge as  $N_{\text{rep}} \rightarrow \infty$  to the true expectations  $\mu_R(t)$ ,  $\mu_G(t)$  and  $C_{RG}(t)$ —which are exactly the solutions of the moment ODEs. With finitely many replicates the SSA should therefore follow the ODE curves up to a statistical error of order  $1/\sqrt{N_{\text{rep}}}$ .

**Edge 2 — Moment ODEs  $\leftrightarrow$  Steady-state sampler (long time).** The stationary distribution is the  $t \rightarrow \infty$  limit of the dynamics, so the moment ODEs must approach a fixed point, which is (12). The steady-state sampler must estimate these same values, with an error decaying as  $\sigma/\sqrt{N_{\text{rep}}}$  by the Central Limit Theorem, where  $\sigma$  is the standard deviation of the stationary distribution. Observing that rate confirms the sampler is a valid Monte Carlo estimator of the steady-state moments.

**Edge 3 — SSA  $\leftrightarrow$  Steady-state sampler (closing the triangle).** An SSA trajectory run long enough to reach stationarity samples the same distribution  $P_{\infty}$  as the direct sampler, so the long-time plateau of the SSA moments must agree with both the ODE fixed point and the sampler estimates. Because both endpoints of this edge produce samples, the claim can be sharpened to equality of the full laws,  $\hat{P}_{\infty}^{\text{SSA}} = \hat{P}_{\infty}^{\text{smp}}$ , rather than of their first two moments alone; Section 4.4 tests it in total-variation distance.

### 4.2 Parameter Exploration

This part runs the SSA on its own to check that it reproduces the expected behaviour when the gene’s parameters change. We vary one mechanism at a time so that each run isolates a single effect

(Table 1). The final run changes numerical settings rather than biological mechanisms, testing the stability of the simulation itself.

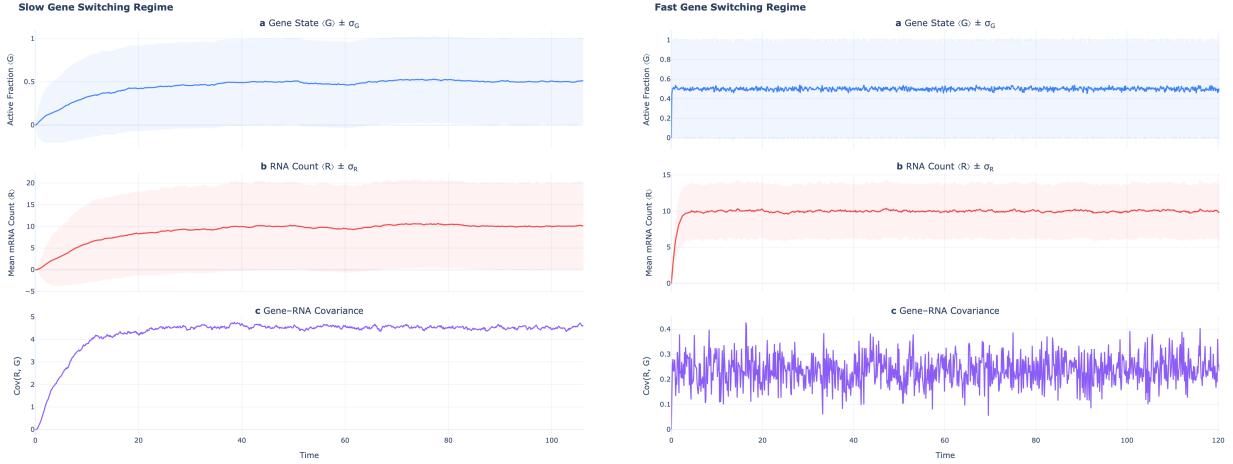


Figure 2: Switching speed reshapes noise at a fixed mean ( $\mu_R \approx 10$ ). Slow switching (left,  $k_{\text{on}} = k_{\text{off}} = 0.05$ ) produces long bursts, a wide  $\pm\sigma_R$  band, and strong gene–RNA covariance. Fast switching (right,  $k_{\text{on}} = k_{\text{off}} = 10$ ) averages the promoter out, leaving a narrow band and near-zero covariance.

Three results emerge from these runs. First, switching speed sets the noise at a fixed mean: as the switching rates grow relative to  $k_{\text{deg}}$  (Figure 2), the gene state averages out,  $\sigma_R^2$  and  $C_{RG}$  fall, and  $F$  drops toward the Poisson baseline without reaching it at any finite rate—the behaviour predicted by the exact limits of [11].

Second, with  $k_{\text{on}}$  fixed, the burst size (2) sets the steady-state mean  $\mu_R^{ss}$ . The expression runs show that a weaker promoter makes fewer molecules but has a larger relative spread (coefficient of variation,  $\sigma_R/\mu_R$ ), which is why genes expressed at low copy number appear noisier.

Third, the steady state depends on neither the initial condition nor the numerical resolution. The two initialisations reach the same Fano factor, as they must for an ergodic chain, and enlarging the ensemble reduces sampling error without moving the steady state. The smaller ensemble is set to the cell counts typical of single-molecule mRNA studies [5].

| Experiment      | Parameters                                                                   | $F^{ss}$ | $\hat{F}$ (SSA) $\pm$ [INSERT_SE] | What it isolates        |
|-----------------|------------------------------------------------------------------------------|----------|-----------------------------------|-------------------------|
| Slow switching  | $k_{\text{on}} = k_{\text{off}} = 0.05$ , $k_{\text{syn}} = 20$              | 10.09    | 9.94                              | noise at fixed mean     |
| Fast switching  | $k_{\text{on}} = k_{\text{off}} = 10$ , $k_{\text{syn}} = 20$                | 1.48     | 1.48                              | noise at fixed mean     |
| High expression | $k_{\text{on}} = 0.5$ , $k_{\text{off}} = 4$ , $k_{\text{syn}} = 56$         | 10.05    | 9.90                              | burst size $\approx 14$ |
| Low expression  | $k_{\text{on}} = 0.5$ , $k_{\text{off}} = 10$ , $k_{\text{syn}} = 30$        | 3.48     | 3.49                              | burst size $\approx 3$  |
| Initial cond.   | $(g_0, r_0) = (0, 0)$ vs. $(1, 15)$                                          | 9.33     | 9.41 / 9.27                       | forgetting the start    |
| Numerical       | $n_{\text{sim}} = 1000$ vs. $N_{\text{rep}} = 224$ , $n_{\text{sim}} = 5000$ | 9.33     | 9.48 / 9.40                       | estimate quality        |

Table 1: The six parameter-exploration runs. Unless shown otherwise,  $k_{\text{deg}} = 1$ , the base switching rates are  $k_{\text{on}} = k_{\text{off}} = 0.1$ , and  $N_{\text{rep}} = [\text{INSERT\_NREP}]$ ; each run reaches the steady-state mean  $\mu_R^{ss} = 10$  except the expression runs, which change it on purpose.  $F^{ss}$  is the closed-form steady-state Fano factor (11);  $\hat{F} = \hat{\sigma}_R^2 / \hat{\mu}_R$  is measured from the corresponding SSA run, averaged over the last fifth of the horizon, and carries a Monte Carlo standard error of [INSERT\_SE]. Rows with two runs report both.



The SSA therefore reproduces the qualitative features expected of the telegraph model, and its measured Fano factors agree with the closed form (11) to within 2% across all six runs—a deviation comparable to the Monte Carlo standard error of each estimate (Table 1).

### 4.3 SSA vs. ODE: Moment Dynamics (Edge 1)

Edge 1 of Section 4.1 asserts that the stochastic simulation and the deterministic moment ODEs describe the same dynamics at all times. The direct test is the overlay in Figure 3. The SSA ensemble averages follow the deterministic ODE curves throughout the transient relaxation and into the steady state. The visible deviations are confined to the covariance curve, the noisiest of the three statistics, and shrink as  $N_{\text{rep}}$  increases. The  $\pm\sigma$  bands are wide, but they measure cell-to-cell variability rather than numerical error, and they match the analytical spread predicted by the moment ODEs.

The agreement is not limited to the symmetric case: the SSA tracks the deterministic dynamics under slow and fast switching, high and low expression, and both initial conditions. Figure 4 shows the least favourable of these, the high-noise slow-switching regime, where the tracking is unchanged.

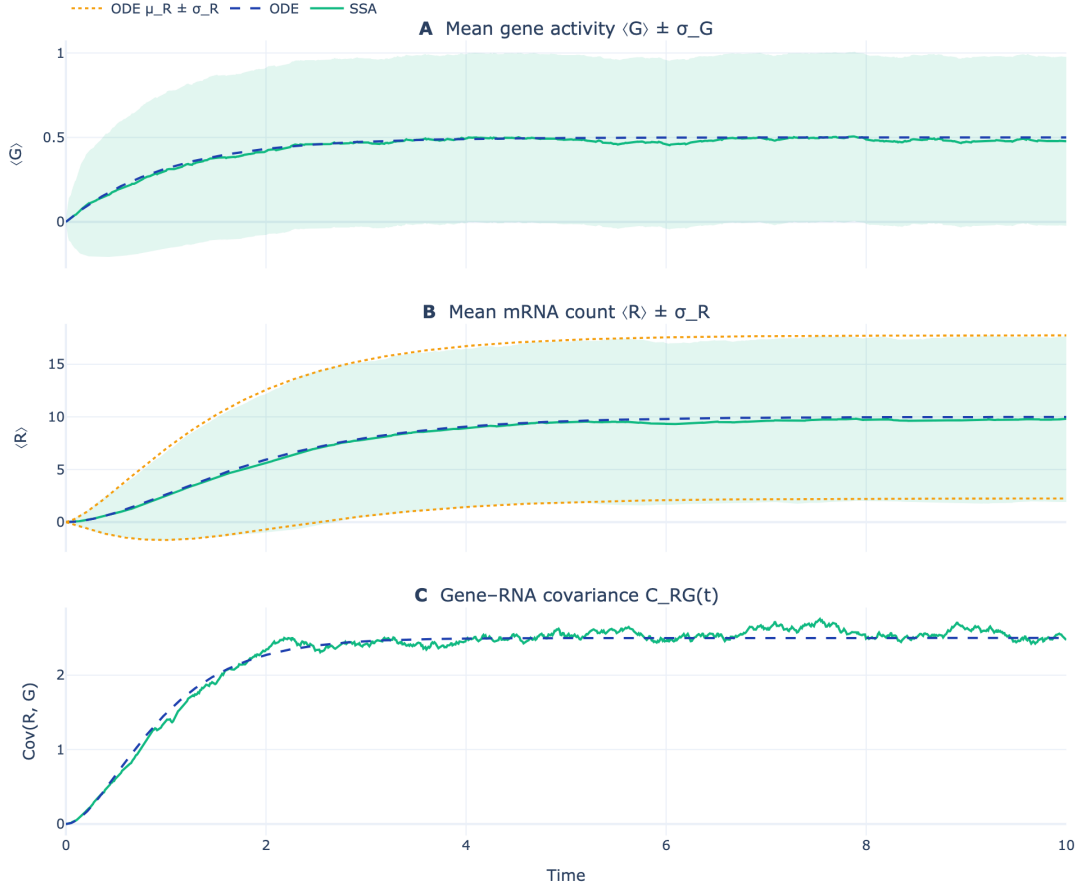


Figure 3: SSA ensemble averages (green, with empirical  $\pm\sigma$  fluctuation bands shaded) and the deterministic ODE solutions (dashed blue) at the reference parameters from  $(g_0, r_0) = (0, 0)$ , with  $N_{\text{rep}} = 1000$ . In the mRNA panel, the analytical spread from the moment ODEs,  $\mu_R \pm \sigma_R$ , is drawn as amber dotted lines on top of the SSA band.

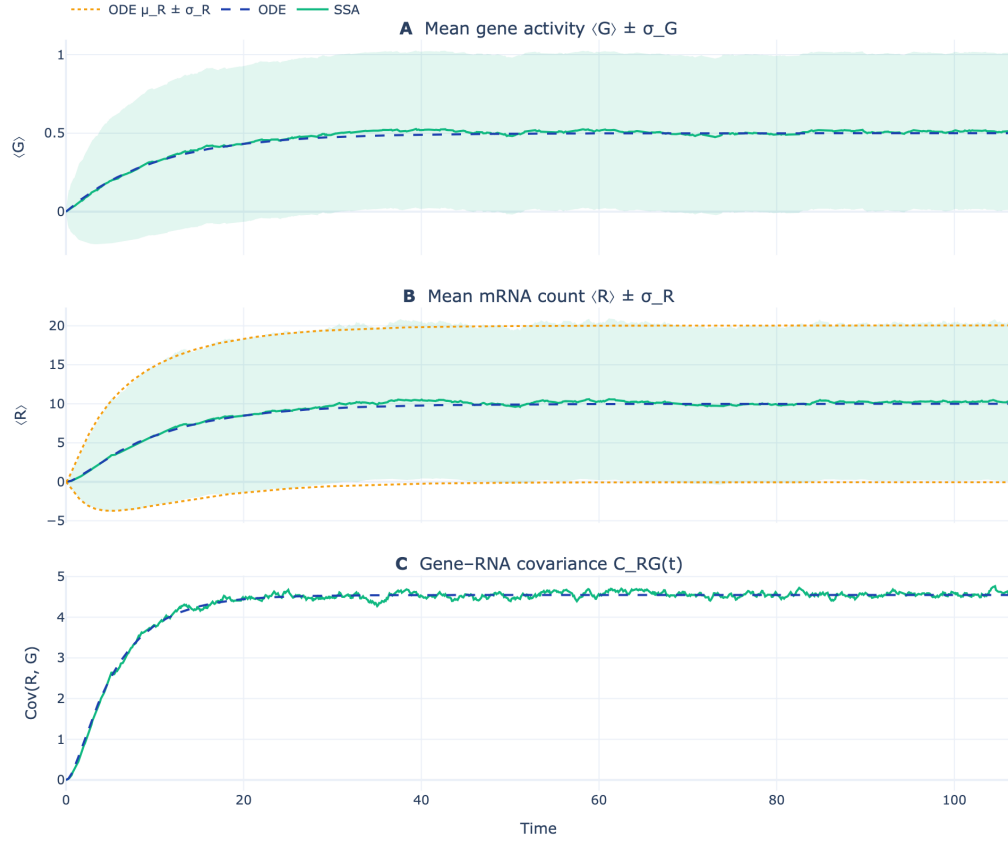


Figure 4: SSA versus moment ODEs in the high-noise slow-switching regime. Formatting follows Figure 3. Even where fluctuations are largest, the SSA ensemble averages track the deterministic ODE curves throughout the transient and into steady state, confirming Edge 1 beyond the symmetric reference case.

An overlay shows agreement but does not measure its rate, which Edge 1 predicts to be  $1/\sqrt{N_{\text{rep}}}$ . Figure 5 measures it as the root-mean-square (RMS) deviation of  $\widehat{\mu}_R(t)$  from the ODE curve over the full time grid. We use the RMS rather than the supremum because it aggregates the deviation across every grid point and therefore has far lower variance as an estimator; the supremum is fixed by a single extreme point and converges at a different rate. On log-log axes the deviation is straight over three decades, with a fitted slope of  $-0.521 \pm [\text{INSERT\_CI}]$  against the  $-1/2$  predicted by the Law of Large Numbers.

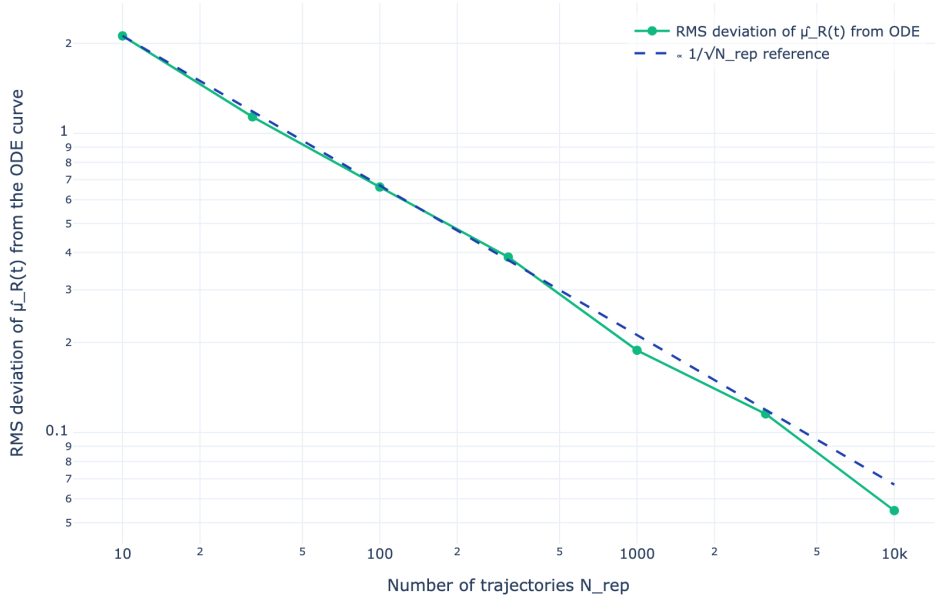


Figure 5: Convergence of the SSA ensemble mean to the moment-ODE solution (Edge 1) at the reference parameters. The statistic is the RMS deviation of  $\widehat{\mu}_R(t)$  from the ODE curve over the full time grid; the dashed line is a  $1/\sqrt{N_{\text{rep}}}$  reference anchored at the smallest ensemble. This is the Edge 1 counterpart of Figure 6, which tests Edge 2.

The SSA therefore converges to the moment dynamics at the rate predicted by the Law of Large Numbers, and Edge 1 holds across every regime tested. What converges is the estimate of the mean; the  $\pm\sigma$  bands measure the persistent cell-to-cell variability described by the variance equation (8) and are independent of  $N_{\text{rep}}$ .

#### 4.4 Steady-State Validation (Edges 2 & 3)

Edges 2 and 3 of Section 4.1 concern the stationary regime. This section tests both, and checks that all three descriptions reduce correctly to the known limits at the boundaries of parameter space.

The ODE solutions in Figure 3 plateau at the analytical fixed points (12), and the direct sampler reproduces the same values to within Monte Carlo error. Figure 6 makes the convergence quantitative: on log-log axes the mean absolute deviation of the sampler’s  $\mu_R^{ss}$  estimate follows a  $-1/2$  slope over four decades.

A stricter check asks the sampler to reproduce *all five* steady-state moments at once. Table 2 compares the sampled moments against the closed forms (17) and (12) in an asymmetric regime. Every deviation is between one and three standard errors, the range expected from sampling noise alone, with no sign of systematic bias.

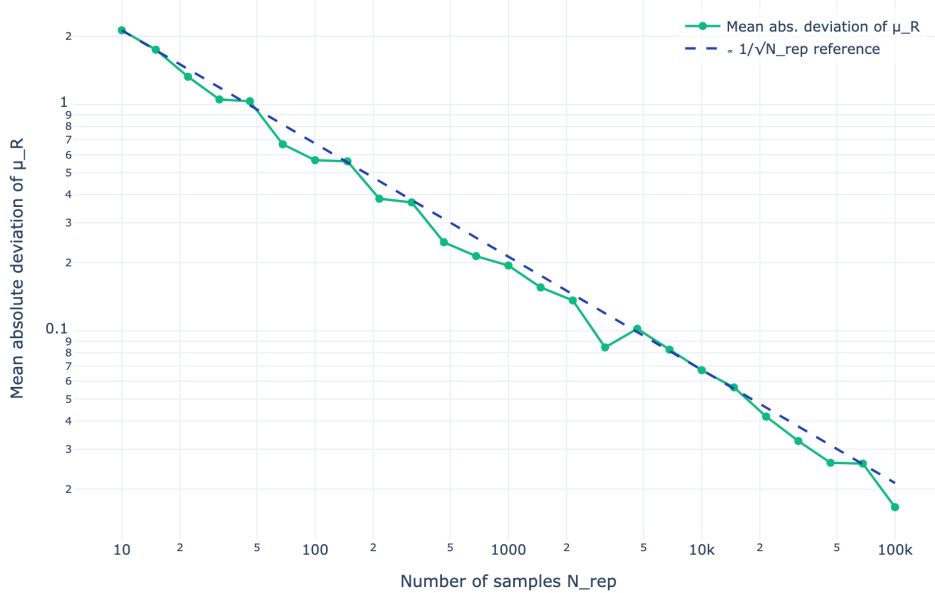


Figure 6: Convergence of the exact steady-state sampler’s empirical moments to the analytical steady-state values as a function of the ensemble size  $N_{\text{rep}}$ . The error (mean absolute deviation) decays as  $1/\sqrt{N_{\text{rep}}}$ , the rate given by the Central Limit Theorem.

| Moment          | Analytic | Sampler | Abs. diff. | Std. error |
|-----------------|----------|---------|------------|------------|
| $\mu_G^{ss}$    | 0.3333   | 0.3317  | 0.0016     | 0.0010     |
| $\mu_R^{ss}$    | 6.6667   | 6.6277  | 0.0389     | 0.0142     |
| $\sigma_G^{ss}$ | 0.4714   | 0.4708  | 0.0006     | 0.0004     |
| $\sigma_R^{ss}$ | 6.4979   | 6.4767  | 0.0212     | 0.0105     |
| $C_{RG}^{ss}$   | 1.7778   | 1.7666  | 0.0111     | 0.0071     |

Table 2: Steady-state moments from the direct sampler ( $N_{\text{rep}} = 2 \times 10^5$ ) against exact analytical values for  $k_{\text{on}} = 0.5, k_{\text{off}} = 1, k_{\text{syn}} = 20, k_{\text{deg}} = 1$ . The final column is the Monte Carlo standard error obtained via bootstrapping. Because all five moments are computed from a single array of Beta draws, their deviations are strongly correlated. For instance, the mean mixing fraction of this specific draw falls  $\approx 0.0016$  low; since  $\mu_R = (k_{\text{syn}}/k_{\text{deg}})\bar{p}$  multiplies this shortfall by 20, it accounts for most of the observed deviation in  $\mu_R^{ss}$ . Thus, one underlying fluctuation drives the differences across multiple rows.

The boundary limits predicted in Section 3.3 also hold. At  $k_{\text{off}} = 0$  the model reduces to a birth–death process with a Poisson steady state [13, 14]: the moment ODEs relax to  $\mu_G^{ss} = 1$  and  $\mu_R^{ss} = k_{\text{syn}}/k_{\text{deg}}$ , and the sampled counts overlay the analytic Poisson PMF (Figure 8) with a measured Fano factor of  $0.99 \pm [\text{INSERT\_SE}]$  against the theoretical 1. At the opposite boundary,  $k_{\text{deg}} = 0$ , mRNA accumulates without bound and no steady state exists.

**Edge 3 as a distributional test.** Two different distributions can share a mean, a variance, and a covariance, so agreement of the first two moments is weaker evidence than it appears. Since both endpoints of Edge 3 produce samples, we can compare the full stationary laws instead, using the Total Variation distance  $\text{TV} = \frac{1}{2} \sum_r |\hat{P}^{\text{SSA}}(r) - \hat{P}^{\text{smp}}(r)|$ . Unlike a  $\chi^2$  test, TV needs no binning of near-empty states, is bounded in  $[0, 1]$ , and is the largest probability by which the two laws can differ on any event. On the SSA side the mRNA count is recorded at one late time point per trajectory, which costs samples but avoids the autocorrelation that would understate the spread.

The null distribution for TV is obtained by *direct resimulation*: assuming both methods sample the same stationary law, we draw 200 fresh pairs at the same sample sizes and recompute the statistic. We avoid bootstrapping each side from its own empirical measure, which would add sampling noise twice and centre the null band above the true expected TV, making the test too lenient.

The slow-switching regime (Figure 7, right) is the demanding case, and the one where moment-matching is least informative. Its stationary law is bimodal, with mass at  $R = 0$  (long OFF periods) and near  $R = k_{\text{syn}}/k_{\text{deg}}$  (long ON periods) and the mean falling in the trough between them. A method could match  $\mu_R$  and  $\sigma_R$  exactly and still miss this shape entirely; the  $\pm\sigma$  bands used elsewhere in this paper suggest a Gaussian profile that this law does not have.

Measured by TV, the two laws agree. In the slow regime  $\text{TV} = 0.0117$  against a null median of 0.0130 (95th percentile 0.0166, one-sided  $p = 0.74$ ); in the symmetric regime  $\text{TV} = 0.0093$  against a median of 0.0155 (95th percentile 0.0191,  $p = 1.00$ ). Neither exceeds what sampling variation alone would produce.

The symmetric case falls below all 200 null draws, hence  $p = 1.00$ . This value indicates that our resimulated null distribution may be conservative, or that the empirical variance matched the analytic expectation closely enough to flatter the test statistic. The second reading is consistent with what we observe: the empirical variance of 59.8 matches the analytic  $\sigma_R^2 = 60$  to within 0.4%, and splitting the sample in half reproduces the dispersion of two independent sampler draws. Calibrating the null across many seeds would separate the two explanations; we did not do so here. Under either reading, Edge 3 compares the hand-written SSA loop against a sampler that follows in closed form from (13).

Taken together, these checks show the sampler to be an unbiased,  $\sqrt{N}$ -consistent estimator of the steady-state moments, so Edge 2 holds. Edge 3 holds independently rather than by inheritance from Edges 1 and 2, and at the level of the full stationary distribution rather than its first two moments.

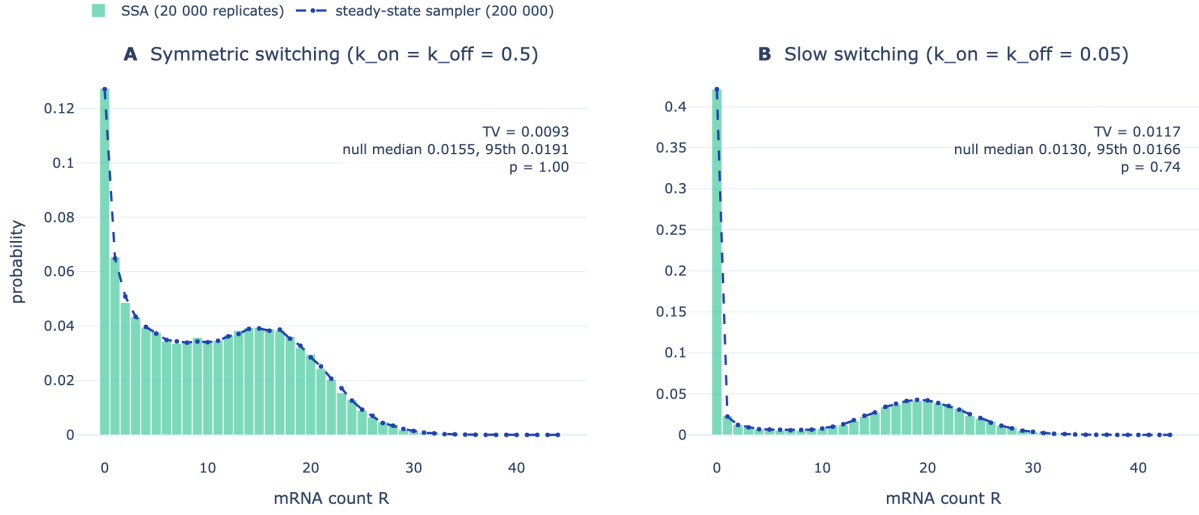


Figure 7: Edge 3 as a distributional test: the stationary mRNA distribution from 20 000 independent SSA trajectories (green bars, one sample per trajectory) against  $2 \times 10^5$  draws from the steady-state sampler (dashed blue). Left: symmetric reference regime. Right: slow switching ( $k_{\text{on}} = k_{\text{off}} = 0.05$ ), where the distribution is bimodal.

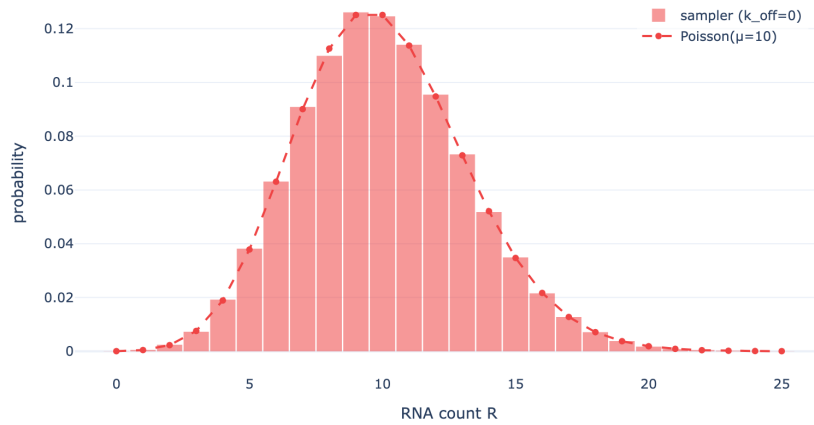


Figure 8: Boundary case  $k_{\text{off}} = 0$  (permanently active gene): the sampled steady-state mRNA distribution against the analytic Poisson PMF with mean  $k_{\text{syn}}/k_{\text{deg}}$ .

## 5 Conclusion

All three edges of the validation triangle hold. Their value lies in what the three implementations do not share: a trajectory simulator, a deterministic ODE integrator, and a Beta–Poisson sampler have no common code and fail in different ways, so an error in the SSA event loop, a mis-derived moment equation, or a wrong stationary law would each disturb one edge while leaving the others intact. The reductions to classical limits at  $k_{\text{off}} = 0$  and  $k_{\text{deg}} = 0$  anchor this at the two boundaries where the answer is known in advance.

The triangle is a validation of the three implementations against one another over the regimes tested, not a proof of correctness over the whole parameter space; and because all three descend from the same master equation (3), it cannot detect an error in that shared starting point. What it provides is a template: any extension of the model can be held to the same three-way standard, and the next section describes two such extensions.

## 6 Future Work

With the telegraph model established as a validated baseline, two extensions would capture more of the underlying biology. A third direction would strengthen the validation itself: a direct numerical solution of the master equation by finite-state projection would supply an external reference for the full time-dependent law, extending the triangle beyond mutual consistency between our own implementations.

**Incorporating the protein level.** The natural next step is a three-stage model (gene  $\rightarrow$  mRNA  $\rightarrow$  protein) that includes translation at rate  $k_{\text{trans}}$  and protein degradation at rate  $k_{\text{deg},p}$ . The tools validated here carry over directly: the SSA handles the two additional reactions unchanged, and the moment ODE system expands to cover protein variance and mRNA–protein covariance.

**Gene regulatory networks.** In vivo, gene products often toggle other genes, forming regulatory networks that can be described by mapping the interacting genes to an *Ising model*, where ON/OFF states act as spins and regulatory links as couplings. The obstacle is the size of the state space: for  $n$  genes the joint distribution spans  $2^n$  promoter configurations, each multiplying the unbounded mRNA and protein spaces, so direct integration gives way to approximate high-dimensional density estimators.

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